

PDE

O.D.E

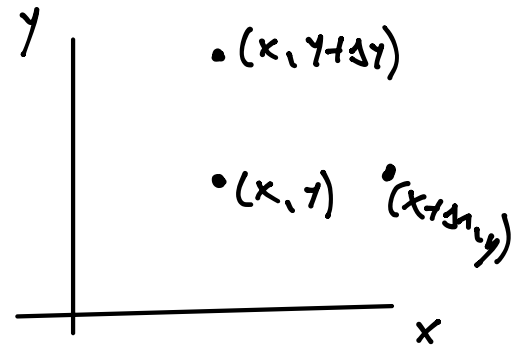
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \quad \text{md}^2 v = f(v, t)$$

Diferencius finitus

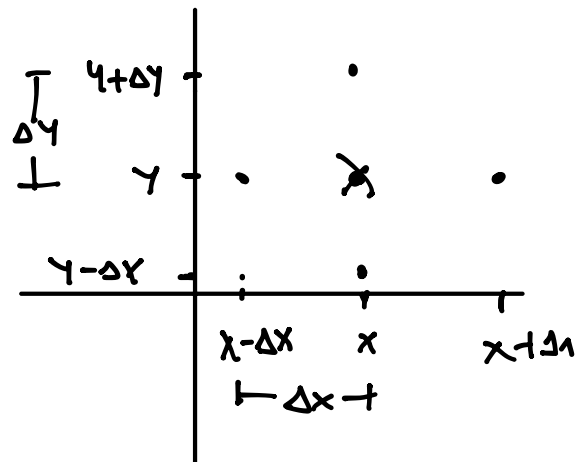
$$\frac{dy}{dx} \approx \frac{\Delta y}{\Delta x}$$

$$u(x, y) \quad \frac{\partial v}{\partial x} \approx \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}$$

$$\frac{\partial v}{\partial y} \approx \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y}$$



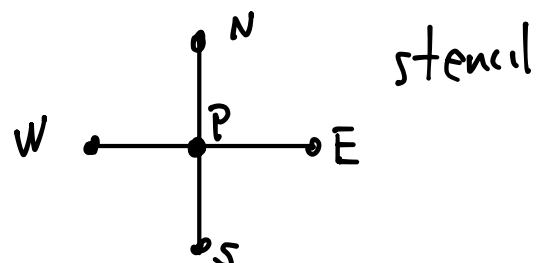
$$\frac{\partial v}{\partial x} \approx \frac{v(x, y) - v(x - \Delta x, y)}{\Delta x}$$



$$\frac{\partial v}{\partial x} \approx \frac{v(x + \Delta x, y) - v(x - \Delta x, y)}{2 \Delta x}$$

$$\frac{\partial v}{\partial y} \approx \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} \quad \frac{\partial v}{\partial y} \approx \frac{v(x, y) - v(x, y - \Delta y)}{\Delta y}$$

$$\frac{\partial v}{\partial y} \approx \frac{v(x, y + \Delta y) - v(x, y - \Delta y)}{2 \Delta y}$$



$$\frac{\partial u}{\partial x} = \frac{U_E - U_P}{\Delta x}$$

$$\frac{\partial u}{\partial x} = \frac{U_P - U_W}{\Delta x}$$

$$\frac{\partial u}{\partial x} = \frac{U_E - U_W}{2\Delta x}$$

$$\frac{\partial u}{\partial y} = \frac{U_N - U_P}{\Delta y}$$

$$\frac{\partial u}{\partial y} = \frac{U_P - U_S}{\Delta y}$$

$$\frac{\partial u}{\partial y} = \frac{U_N - U_S}{2\Delta y}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{U_E - 2U_P + U_W}{\Delta x^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{U_N - 2U_P + U_S}{\Delta y^2}$$

Case	Description	Examples of explicit finite difference formulae	Error level
1	Centered, regular grid	$f''(a) \approx \left[-\frac{1}{12} f(a-2h) + \frac{4}{3} f(a-h) - \frac{5}{2} f(a) \right. \\ \left. + \frac{4}{3} f(a+h) - \frac{1}{12} f(a+2h) \right] / h^2$	$O(h^4)$
2	Staggered, regular grid	$f'(a) \approx \left[\frac{1}{24} f(a-\frac{3}{2}h) - \frac{9}{8} f(a-\frac{1}{2}h) \right. \\ \left. + \frac{9}{8} f(a+\frac{1}{2}h) - \frac{1}{24} f(a+\frac{3}{2}h) \right] / h$	$O(h^4)$

Examples of explicit finite difference formulae

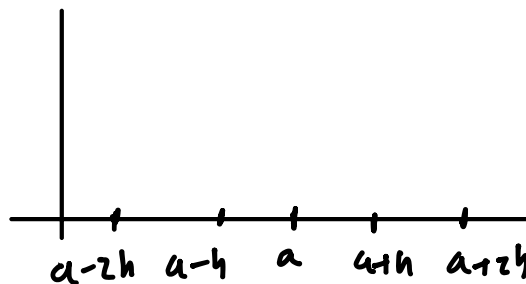


Table 6.3 Central-Difference Formulas of Order $O(h^2)$

$$f'(x_0) \approx \frac{f_1 - f_{-1}}{2h}$$

$$f''(x_0) \approx \frac{f_1 - 2f_0 + f_{-1}}{h^2}$$

$$f^{(3)}(x_0) \approx \frac{f_2 - 2f_1 + 2f_{-1} - f_{-2}}{2h^3}$$

$$f^{(4)}(x_0) \approx \frac{f_2 - 4f_1 + 6f_0 - 4f_{-1} + f_{-2}}{h^4}$$

